

1 Exact third growth on the moving second holding state

The third growth begins at the actual second return. Its inherited first vorticity is positive, its second vorticity is zero, and its second temperature and phase length continue changing. We derive the complete third phase and linear pair on the holding interval already proved in Theorem 2. The original source growth control is $\dot{\alpha} = \mathcal{F}_2$; the different control $\mathcal{F}_3$ is retained below as a computed function. This distinction matters at the next transition. All formulas below use the physical frequency $K_j = \mu\hat{\lambda}_j$, the original $\nu = \nu_{AB} = \sqrt{A_0}$, and the actual changed scales.

1.1 Actual birth data and their branches

Set $t_0 = t_2^b$, $\sigma_1 = \sigma$ and $s_2 = L_2\nu/\sigma_1$, with $a = \sin s_2$ as in (97). Write $h_b = h(t_0)$, $\chi_b = \sqrt{h_b^2 + a^2}$, $P_2 = -\Theta_2$ and $A_1 = -K_1\Theta_1$. On the proved interval

$$I = \left[t_0, t_0 + \frac{1}{32\Gamma_2} \right], \quad \Gamma_2 = \sigma_1 a, \quad (1)$$

the actual physical holding equations and phases are

$$\begin{aligned} \chi &= \sqrt{h^2 + a^2}, \quad \zeta_1 = \frac{(-a, h)}{\chi}, \quad \zeta_2 = (0, \chi), \quad \theta = \arctan(a/h), \quad \alpha = s_* + \theta, \\ \dot{h} &= a\Omega_1, \quad \dot{\Omega}_1 = \frac{aA_1}{\chi} - dK_1^2\Omega_1, \quad \dot{A}_1 = -A_0 \sin s_*\Omega_1 - dK_1^2A_1, \\ \Omega_2 &= 0, \quad \dot{P}_2 = -dK_2^2\chi^2P_2, \quad \dot{\alpha} = -\frac{a^2\Omega_1}{\chi^2}. \end{aligned} \quad (2)$$

Here A_1, P_2, Ω_1 are strictly positive throughout I ; $h_b \leq h < 3$, $h_b \geq \cos s_2 \geq 15/16$ and $1 \leq \chi < \sqrt{10}$ are proved bounds, not frozen values. The notation θ in this subsection denotes an angle only; the temperature field is denoted ϑ below where needed.

Retain the original recursion

$$\begin{aligned} Q_3 &= Q_* + 3, \quad \hat{\lambda}_3 = \hat{\lambda}_2^{Q_3}, \quad k_3 = \max\{k \in \mathbb{N}_0 : 120k \leq Q_3, K(k) \leq \log \hat{\lambda}_3\}, \\ L_3 &= (k_3 + 41/8) \log \hat{\lambda}_3, \quad \sigma_2^2 = |G_{<3}(t_0)|, \quad s_3 = \frac{L_3\sigma_1}{\sigma_2}, \quad \Gamma_3 = \sigma_2 \sin s_3, \quad p_3 = e(s_3). \end{aligned} \quad (3)$$

The positive square root defining σ_2 is the source scale evaluated on the actual state. It is not the magnitude of only the second temperature contribution. The set defining k_3 is nonempty because $k = 0$ belongs to it by (72), and finite because $120k \leq Q_3$.

For completeness, even the earlier frequency choice fixes the small positive branch needed for the exact map. Indeed $\alpha \leq 1/4$ on I , so the vertical component of $G_{<3} = -A_0R_\alpha e_2 - A_1\zeta_1 - K_2P_2\zeta_2$ at birth has three negative summands. Therefore

$$\sigma_2^2 \geq K_2P_2(t_0)\chi_b \geq A_0\hat{\lambda}_2^{1/8}, \quad \sigma_2 \geq \nu\hat{\lambda}_2^{1/16}.$$

The threshold and selected gain in Theorem 1 justify the second inequality. Put $y = \log \hat{\lambda}_1$. Equations (79) and (72) give $\sigma_1 \leq C_\sigma \nu e^{y/16}$ and $y \geq Q_2 \geq 202$. Since $\Lambda_1 \geq 6$, one has $C_\sigma = \sqrt{1 + e^{2+\Lambda_1}^{-1}} < 6$. One may verify the numerical bound by $e < 3$, which follows from $n! \geq 2^{n-1}$ for $n \geq 2$ and strict inequality for $n \geq 3$. Also $Q_3 = Q_2 + 1 \leq 2Q_2$ and $k_3 + 41/8 \leq Q_3/120 + 6 \leq Q_3$; hence $L_3 \leq 8Q_2^3y$. Consequently

$$0 < s_3 \leq 48Q_2^3ye^{-(Q_2-1)y/16} \leq 48y^4e^{-12y} \leq \frac{5760}{12^5y} < \frac{1}{8}. \quad (4)$$

The penultimate inequality is the fifth positive term in the exponential series. Thus $0 < \theta_b := \arctan(a/h_b) \leq s_2 < 1/8$ and $0 < s_3 + \theta_b < 1/4$. These are statements about the original source

angle, including its branch. Stronger angle and physical diffusion bounds needed to reach the third threshold are supplied by the accompanying actual third-growth theorem; none is presumed in the identities below.

Define the oriented constants and the changing inner product by

$$b = \sin(s_3 + \theta_b), \quad g_b = \cos(s_3 + \theta_b), \quad g = g_b + \frac{b}{a}(h - h_b), \quad c = bh_b - ag_b = \chi_b \sin s_3 > 0. \quad (5)$$

The constant b in (5) is dimensionless and must not be confused with the physical buoyancy coefficient $\mathfrak{b}_3$ below. One has $b > 0$, $g_b > 15/16$, $g \geq g_b$, and the exact identity $bh - ag = c$ on I .

1.2 The complete material map, phases and original feedback

For $v \in \mathbb{R}^2$, the matrix convention is $Jv = (-v_2, v_1)$ and $R_\beta = \exp(\beta J)$. The third deformation and its inverse are

$$\begin{aligned} M_3(t) &= M_2(t)M_2(t_0)^{-1} = R_\theta \begin{pmatrix} 1 & -(h - h_b)/a \\ 0 & 1 \end{pmatrix} R_{-\theta_b}, \\ M_3(t)^{-1} &= R_{\theta_b} \begin{pmatrix} 1 & (h - h_b)/a \\ 0 & 1 \end{pmatrix} R_{-\theta}. \end{aligned} \quad (6)$$

These formulas describe linear maps $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ in the original physical coordinates and have determinant one. In particular $M_3(t_0) = I$ retains the source third-birth datum; it does not reset M_1 or M_2 . Their phase pullback gives

$$\begin{aligned} \zeta_3 &= M_3^{-T} p_3 = g\zeta_1 - bJ\zeta_1 = \frac{(c, hg + ab)}{\chi}, \quad R_3 := |\zeta_3|^2 = g^2 + b^2, \\ \det(\zeta_1, \zeta_3) &= -b, \quad \det(\zeta_2, \zeta_3) = -c, \quad \dot{g} = b\Omega_1, \quad \dot{R}_3 = 2bg\Omega_1, \quad \dot{\chi} = \frac{ah\Omega_1}{\chi}. \end{aligned} \quad (7)$$

Thus $R_3(t_0) = 1$ and $R_3 \geq 1$. Its first component is positive and decreases on growth; it is not held constant by the preceding feedback. The lower component is positive and satisfies the explicit bound

$$\zeta_{3,2} = \frac{hg + ab}{\chi} > \frac{225}{256\sqrt{10}} > \frac{1}{4}. \quad (8)$$

For example the last strict comparison follows on squaring from $225^2 > 64^2 \cdot 10$.

Proof of the geometric map. On holding $\Omega_2 = 0$ and the older cutoffs equal one, so $D_{<3} = D_{<2} = \dot{\alpha}J + \Omega_1 J\zeta_1 \otimes \zeta_1$. The previously proved formula for M_2 is $R_\theta \begin{pmatrix} 1 & -(h - \cos s_2)/a \\ 0 & 1 \end{pmatrix}$. Multiplication by its birth inverse gives (6). Differentiating $M_2(t)M_2(t_0)^{-1}$ gives $\dot{M}_3 = D_{<3}M_3$, including the original initial value. Multiplying the proposed inverse by M_3 on either side gives I .

Since $R_{-\theta_b}e(s_3) = (b, g_b)$, inverse transposition of the shear sends this vector to (b, g) . Subsequent rotation gives $R_\theta(b, g) = (-ag + bh, hg + ab)/\chi$, proving the pullback in (7). Because $\zeta_1 = R_\theta e_2$ and $J\zeta_1 = R_\theta(-e_1)$, this is also $g\zeta_1 - bJ\zeta_1$. Its norm, inner product and signed determinants follow by the displayed orthonormal basis and the two laboratory components. Differentiating $bh - ag = c$ and using $\dot{h} = a\Omega_1$ gives $\dot{g} = b\Omega_1$. Direct differentiation gives the remaining rates. All maps are invertible in their specified domains because $a, h, \chi > 0$; their branch choices were proved above. The lower bound (8) follows from $h, g > 15/16$ and $\chi < \sqrt{10}$. $\square$

Write the original feedback functions at the actual state as

$$\mathcal{F}_2 = -\frac{a^2\Omega_1}{\chi^2}, \quad \mathcal{F}_3 = -\frac{ab\Omega_1}{hg + ab}, \quad \mathcal{F}_3 - \mathcal{F}_2 = -\frac{ach\Omega_1}{\chi^2(hg + ab)}. \quad (9)$$

To verify them from the original sum, use

$$\mathcal{F}_j = -\frac{\sum_{i < j} \Omega_i e_{i,1} (J e_i \cdot \zeta_j)}{\zeta_{j,2}}.$$

Here $e_1 = \zeta_1$, $J\zeta_1 \cdot \zeta_2 = -a$, $J\zeta_1 \cdot \zeta_3 = -b$ and $\Omega_2 = 0$. The two substitutions give the first two formulas in (9). Taking their difference and using $bh - ag = c$ gives the third. In particular the physical controlled-component identity is exactly

$$\dot{\zeta}_{3,1} = (\mathcal{F}_3 - \mathcal{F}_2)\zeta_{3,2} = -\frac{ach\Omega_1}{\chi^3} = \frac{d}{dt} \left(\frac{c}{\chi} \right).$$

No term has been discarded by replacing the growth feedback with the third feedback. Conversely, substituting (9) in the full phase equation reconstructs this same growth map.

1.3 Both inherited temperatures and the exact numerator identity

Put $S_* = A_0 \sin s_*$ and $C_* = A_0 \cos s_*$. These symbols retain the original constants; they are not new adjustable scales. The complete parent gradient can be written in two identical forms:

$$\begin{aligned} G_{<3} &= -A_0 R_\alpha e_2 - A_1 \zeta_1 - K_2 P_2 \zeta_2 \\ &= -(A_1 + C_* + K_2 P_2 h) \zeta_1 - (S_* - a K_2 P_2) J \zeta_1. \end{aligned} \quad (10)$$

The second form follows from $R_\alpha e_2 = \cos s_* \zeta_1 + \sin s_* J \zeta_1$ and $\zeta_2 = h \zeta_1 - a J \zeta_1$. It also supplies the exact source scale formula

$$\sigma_2^4 = (A_1(t_0) + C_* + K_2 P_2(t_0) h_b)^2 + (S_* - a K_2 P_2(t_0))^2. \quad (11)$$

There is no sign restriction on the second bracket, and none is needed in this identity.

Define

$$N_3 = b(A_1 + C_*) + g S_* + c K_2 P_2, \quad E_3 = b K_1^2 A_1 + c K_2^3 \chi^2 P_2. \quad (12)$$

Every summand of N_3 and E_3 is positive on I . The exact scalar-temperature and vorticity couplings are

$$\mathfrak{a}_3 = -\frac{J \zeta_3 \cdot G_{<3}}{K_3 R_3} = \frac{N_3}{K_3 R_3} > 0, \quad \mathfrak{b}_3 = K_3 \zeta_{3,1} = \frac{K_3 c}{\chi} > 0. \quad (13)$$

Their exact rates, including all changing-parent contributions, are

$$\begin{aligned} \dot{N}_3 &= -d E_3, \quad N_3(t) = N_3(t_0) - d \int_{t_0}^t \{b K_1^2 A_1(r) + c K_2^3 \chi(r)^2 P_2(r)\} dr, \\ \frac{d}{dt} \log \mathfrak{a}_3 &= -\frac{d E_3}{N_3} - \frac{2bg\Omega_1}{R_3}, \quad \frac{d}{dt} \log \mathfrak{b}_3 = -\frac{ah\Omega_1}{\chi^2}. \end{aligned} \quad (14)$$

In particular the numerator would be exactly constant at $d = 0$ on this holding geometry. With positive physical diffusivity its replacement is the integral in (14), with both older temperature losses retained. Its positivity follows from the original sum in (12), not from assuming that this integral is small.

Proof. Using (10) and $J\zeta_3 = gJ\zeta_1 + b\zeta_1$ gives

$$-J\zeta_3 \cdot G_{<3} = b(A_1 + C_* + K_2 P_2 h) + g(S_* - a K_2 P_2) = N_3.$$

This proves the full coupling and its signs. Next differentiate the three summands of N_3 , using (2) and (7). The two terms $-bS_*\Omega_1$ and $\dot{g}S_* = bS_*\Omega_1$ cancel exactly; the surviving terms are $-dbK_1^2 A_1$ and $-dcK_2^3 \chi^2 P_2$. Thus $\dot{N}_3 = -dE_3$. Integrating proves its displayed integral identity. Finally apply the logarithmic derivative to (13) and insert $\dot{R}_3$, $\dot{\chi}$ and $\dot{N}_3$. All logarithms are defined because their arguments have already been proved positive. $\square$

1.4 Exact third amplitudes, inverse maps and fundamental functions

The original signed seed and activation are

$$\begin{aligned} S_3 &= \frac{A_0}{\mu} \hat{\lambda}_3^{-k_3-6}, \quad u_i = \sqrt{\mathfrak{b}_3(t_0)/\mathfrak{a}_3(t_0)}, \\ \Theta_3(t_0) &= -S_3, \quad \Omega_3(t_0) = -S_3 u_i, \quad w_3(t) = \eta(\Gamma_3(t - t_0)). \end{aligned} \quad (15)$$

The homogeneous pair is active in the coefficient construction at t_0 , while its physical increment is multiplied by w_3 , exactly as in source (3.11). The full physical equal-diffusion equations are

$$\dot{\Theta}_3 = \mathfrak{a}_3 \Omega_3 - dK_3^2 R_3 \Theta_3, \quad \dot{\Omega}_3 = \mathfrak{b}_3 \Theta_3 - dK_3^2 R_3 \Omega_3. \quad (16)$$

Here are convergent fundamental functions for the changing coefficients, so the solution does not rely on a constant-coefficient surrogate. Let

$$\begin{aligned} \mathcal{A}(t) &= \begin{pmatrix} 0 & \mathfrak{a}_3(t) \\ \mathfrak{b}_3(t) & 0 \end{pmatrix}, \quad \mathcal{H}_0(t) = I, \quad \mathcal{H}_n(t) = \int_{t_0}^t \mathcal{A}(r) \mathcal{H}_{n-1}(r) dr, \\ \mathcal{H}(t) &= \sum_{n=0}^{\infty} \mathcal{H}_n(t), \quad \Lambda_3(t) = dK_3^2 \int_{t_0}^t R_3(r) dr, \quad \mathcal{U}(t) = e^{-\Lambda_3(t)} \mathcal{H}(t). \end{aligned} \quad (17)$$

Each term is an explicit ordered integral of the actual coefficients. The solution and inverse-data map are

$$\begin{aligned} \begin{pmatrix} \Theta_3(t) \\ \Omega_3(t) \end{pmatrix} &= -S_3 \mathcal{U}(t) \begin{pmatrix} 1 \\ u_i \end{pmatrix}, \quad \begin{pmatrix} \Theta_3(t_0) \\ \Omega_3(t_0) \end{pmatrix} = e^{\Lambda_3(t)} \mathcal{H}(t)^{-1} \begin{pmatrix} \Theta_3(t) \\ \Omega_3(t) \end{pmatrix}, \\ \det \mathcal{U} &= e^{-2\Lambda_3} > 0. \end{aligned} \quad (18)$$

Existence, inversion and signs. All parent coefficients are smooth on the compact interval I , and their denominators have the positive bounds already proved. For any matrix operator norm put $M = \max_I \|\mathcal{A}\|$ and $\ell = |I|$. Repeated integration proves $\|\mathcal{H}_n(t)\| \leq M^n (t - t_0)^n / n!$ by induction. The scalar exponential series therefore proves absolute uniform convergence. Also $\mathcal{H}'_n = \mathcal{A} \mathcal{H}_{n-1}$ for $n \geq 1$; the same bound with index $n - 1$ proves uniform convergence of the derivative series. Termwise differentiation gives $\mathcal{H}' = \mathcal{A} \mathcal{H}$ and $\mathcal{H}(t_0) = I$. If two fundamental solutions had the same data, their difference V would obey $V(t) = \int_{t_0}^t \mathcal{A}(r) V(r) dr$. Iteration bounds its norm by $\sup_I \|V\| (M\ell)^n / n!$ for every n , which tends to zero; this proves uniqueness without assuming invertibility first.

For $\mathcal{H} = \begin{pmatrix} f_1 & f_2 \\ g_1 & g_2 \end{pmatrix}$, differentiation of $f_1 g_2 - f_2 g_1$ with $\dot{f}_j = \mathfrak{a}_3 g_j$, $\dot{g}_j = \mathfrak{b}_3 f_j$ gives zero. Its initial value is one. Thus its inverse is exactly $\begin{pmatrix} g_2 & -f_2 \\ -g_1 & f_1 \end{pmatrix}$. Differentiating $\mathcal{U}$ now proves the full damped matrix equation; multiplying by e^{Λ_3} proves its inverse and the determinant in (18). Every entry in $\mathcal{A}$ is nonnegative, and its two off-diagonal entries are positive. Consequently every ordered-integral term has nonnegative entries, the diagonal of $\mathcal{H}$ is at least one, and its off-diagonal entries are positive for $t > t_0$ because already the first term has positive such entries. Since $S_3, u_i > 0$, both amplitudes in (18) are strictly negative throughout I . Hence the temperature sign and all following quotients are established on the complete interval. $\square$

An equivalent scalar fundamental equation retains the geometry in a particularly direct form. Put $Y = e^{\Lambda_3} \Omega_3$. Then

$$\frac{d}{dt}(\chi \dot{Y}) - \frac{cN_3}{R_3} Y = 0, \quad \Omega_3 = e^{-\Lambda_3} Y, \quad \Theta_3 = e^{-\Lambda_3} \frac{\chi \dot{Y}}{K_3 c}. \quad (19)$$

Indeed the first transformed equation is $\dot{X} = \mathfrak{a}_3 Y$, where $X = e^{\Lambda_3} \Theta_3$, and the second is $\dot{Y} = (K_3 c / \chi) X$. The latter gives $\chi \dot{Y} = K_3 c X$; differentiating it and substituting the former gives

(19). Conversely that scalar equation and its displayed inverse give both transformed first-order equations, and hence (16). In physical damped variables the same operator is exactly

$$(\partial_t + \lambda_3)\{\chi(\partial_t + \lambda_3)\Omega_3\} - \frac{cN_3}{R_3}\Omega_3 = 0, \quad \lambda_3 = dK_3^2 R_3. \quad (20)$$

Its complete expanded left side is

$$\chi\ddot{\Omega}_3 + (\dot{\chi} + 2\chi\lambda_3)\dot{\Omega}_3 + (\chi\dot{\lambda}_3 + \dot{\chi}\lambda_3 + \chi\lambda_3^2 - cN_3/R_3)\Omega_3.$$

In particular $\dot{\lambda}_3 = 2dK_3^2 bg\Omega_1$ is present in the physical second-order equation. The integrating factor cancels the newest diagonal damping on the same evolved parent; it does not replace N_3, h, P_2 by their inviscid counterparts.

1.5 Exact quotient, gain and diffusion accounting

Set $\rho = \Gamma_3(t - t_0)$, $v_3 = \sigma_2\Omega_3/(K_3\Theta_3)$ and $P_3 = -\Theta_3$. This change has inverse $t = t_0 + \rho/\Gamma_3$, $\Theta_3 = -P_3$, $\Omega_3 = -K_3P_3v_3/\sigma_2$. Define

$$z_3 = \frac{\chi b}{\chi}, \quad k_3^{\text{dyn}} = \frac{N_3}{\sigma_2^2 \sin s_3 R_3}, \quad \delta_3 = \frac{dK_3^2}{\Gamma_3}. \quad (21)$$

The symbol k_3^{dyn} is a time-dependent coupling and is distinct from the original integer k_3 in (3). Direct quotient and product differentiation prove the exact mutually inverse equations

$$\frac{dv_3}{d\rho} = z_3 - k_3^{\text{dyn}}v_3^2, \quad \frac{d}{d\rho} \log P_3 = k_3^{\text{dyn}}v_3 - \delta_3 R_3, \quad v_3(0) = \frac{\sigma_2 u_i}{K_3}. \quad (22)$$

Thus z_3 is the changing preceding phase-length ratio, not a prescribed constant control. The direct diffusion term cancels from the quotient only because both physical diffusivities equal d . For unequal physical diffusivities κ and ε in the newest pair on this same parent, the corresponding exact term is $(\kappa - \varepsilon)K_3^2 R_3 v_3/\Gamma_3$, and the logarithmic loss is $\kappa K_3^2 R_3/\Gamma_3$. These are algebraic maps for that pair; they make no assertion that the older equal-diffusion trajectory solves the unequal-diffusion parent equations.

The instantaneous eigenline formulation used by the source is also exact. With

$$\gamma_3 = \sqrt{\mathbf{a}_3 \mathbf{b}_3} = \sqrt{\frac{cN_3}{\chi R_3}}, \quad u_* = \sqrt{\mathbf{b}_3/\mathbf{a}_3} = K_3 \sqrt{\frac{cR_3}{\chi N_3}}, \quad Z = \frac{\Omega_3}{u_* \Theta_3},$$

one has, in physical time,

$$\begin{aligned} \dot{Z} &= \gamma_3(1 - Z^2) - Z \frac{d}{dt} \log u_*, & \frac{d}{dt} \log u_* &= \frac{bg\Omega_1}{R_3} - \frac{ah\Omega_1}{2\chi^2} + \frac{dE_3}{2N_3}, \\ \frac{d}{dt} \log P_3 &= \gamma_3 Z - dK_3^2 R_3. \end{aligned} \quad (23)$$

These follow by dividing (16), substituting (14), and differentiating u_* ; the seed has exactly $Z(t_0) = 1$.

For every $t \in I$ the full new-layer diffusion exponent is

$$\begin{aligned} \Lambda_3(t) &= dK_3^2 \int_{t_0}^t \left\{ \left[g_b + \frac{b}{a}(h(r) - h_b) \right]^2 + b^2 \right\} dr, \\ dK_3^2(t - t_0) &\leq \Lambda_3(t) \leq dK_3^2 R_{3,\max}(t - t_0), \end{aligned} \quad (24)$$

where the finite explicit bound is $R_{3,\max} = [g_b + (b/a)(3 - h_b)]^2 + b^2$. The inequalities follow from $h_b \leq h < 3$ and $b, g_b > 0$. The exact gain is the integral

$$\log \frac{P_3(t)}{S_3} = \int_{t_0}^t \gamma_3(r) Z(r) dr - \Lambda_3(t).$$

The two inherited diagonal damping exponents on this same interval are

$$dK_1^2(t - t_0), \quad dK_2^2 \int_{t_0}^t \chi(r)^2 dr.$$

The first temperature also retains the exact response

$$A_1(t) = e^{-dK_1^2(t-t_0)} A_1(t_0) - S_* \int_{t_0}^t e^{-dK_1^2(t-r)} \Omega_1(r) dr.$$

These expressions retain all three physical frequencies and the phase-length factors. The formula for N_3 records their combined effect on the third gain coefficient. No attainment of the third threshold is inferred from positivity of the cooperative transformed pair alone; actual attainment uses the quantitative estimates proved in the accompanying continuation.

1.6 The exact affine parent equations during third growth

The moving affine parent also has an exact spatial interpretation throughout I . Put

$$D = D_{<3}, \quad B_1 = -A_1 \zeta_1, \quad B_2 = -K_2 P_2 \zeta_2, \quad \vartheta(x, t) = G_{<3}(t) \cdot x, \quad u(x, t) = D(t)x.$$

Then

$$\begin{aligned} \dot{G}_{<3} + D^T G_{<3} &= -dK_1^2 B_1 - dK_2^2 \chi^2 B_2, \quad C := D_{21} - D_{12} = 2\dot{\alpha} + \Omega_1, \\ \dot{C} - G_{<3,1} &= 2\ddot{\alpha} - A_0 \sin \alpha - dK_1^2 \Omega_1. \end{aligned} \tag{25}$$

Thus the full equations with equal physical diffusion are satisfied by the explicitly specified scalar force, vector force and pressure

$$\begin{aligned} f_\vartheta &= (-dK_1^2 B_1 - dK_2^2 \chi^2 B_2) \cdot x, \\ f_u &= \frac{1}{2}(2\ddot{\alpha} - A_0 \sin \alpha - dK_1^2 \Omega_1) Jx, \\ p &= -\frac{1}{2} x^T \text{sym}(\dot{D} + D^2 - e_2 \otimes G_{<3})x. \end{aligned} \tag{26}$$

To verify the new temperature contribution, differentiate B_2 using $\dot{P}_2 = -dK_2^2 \chi^2 P_2$ and $\dot{\zeta}_2 = -D^T \zeta_2$; this gives $\dot{B}_2 + D^T B_2 = -dK_2^2 \chi^2 B_2$. The base-plus-first contribution is the exact identity proved in Section 5, so their sum proves the first equation in (25). The curl identity is unchanged because $\Omega_2 = 0$ and $B_{2,1} = 0$, as the displayed vertical phase shows. The trace of D is zero, so D^2 is scalar. Substitution of the quadratic pressure cancels the symmetric part of $\dot{D} + D^2 - e_2 \otimes G_{<3}$; its skew part is $(\dot{C} - G_{<3,1})J/2$. This proves the entire vector equation. Both affine Laplacians vanish by second differentiation. These forces are the affine-core parent forces. The compact released profiles add their exact profile, envelope and interaction terms; the present identities do not discard those spatial residuals or claim an unforced third-layer superposition.